

Figure 14.3 A passing gravitational wave stretches an object one way and then the other. (In this image, the scale of distortion for a typical gravitational wave is hugely exaggerated.)

Of course, we're not particularly interested in detecting gravitational waves produced by nuclear weapons, but the situation with astrophysical sources is not much easier. The closer and more massive the astrophysical source and the more energetic and violent the motion involved, the stronger the gravitational waves we would receive. But even if a star at a distance of 10,000 light-years were to go supernova, as the resulting gravitational wave passed by earth it would stretch a one-meter-long rod by only a millionth of a billionth of a centimeter, barely a hundredth the size of an atomic nucleus. So, unless some highly unexpected astrophysical event of truly cataclysmic proportions were to happen relatively nearby, detecting a gravitational wave will require an apparatus capable of responding to fantastically small length changes.

The scientists who designed and built the *Laser Interferometer GravitationalWave Observatory* (*LIGO*) (being run jointly by the California Institute of Technology and the Massachusetts Institute of Technology and funded by the National Science Foundation) have risen to the challenge. LIGO is impressive and the expected sensitivity is astounding. It consists of two hollow tubes, each *four kilometers* long and a bit over a meter wide, which are arranged in a giant L. Laser light simultaneously shot down vacuum tunnels inside each tube, and reflected back by highly polished mirrors, is used to measure the relative length of each to fantastic accuracy. The idea is that should a gravitational wave roll by, it will stretch one tube relative to the other, and if the stretching is big enough, scientists will be able to detect it.

The tubes are long because the stretching and compressing accomplished by a gravitational wave is cumulative. If a gravitational